

Daily Integral - Medium

$$\int_1^{\infty} \frac{\sinh^{-1}(x)}{x^2} dx$$

This problem was generously submitted by Jordan H. on <https://dailyintegral.com>.

Solution

First, we will integrate by parts. Take $u = \sinh^{-1}(x)$ and $dv = 1/x^2 dx$. Then, $v = -1/x$ and $du = 1/\sqrt{x^2 + 1} dx$. Hence, we get

$$-\frac{\sinh^{-1}(x)}{x} + \int \frac{1}{x\sqrt{x^2 + 1}} dx \Bigg|_1^{\infty}.$$

Now, to evaluate the remaining integral, we use a trigonometric substitution and take $x = \tan(\theta)$. Then, $dx = \sec(\theta)^2 d\theta$. Hence, we have

$$\begin{aligned} & -\frac{\sinh^{-1}(x)}{x} + \int \frac{\sec(\theta)^2}{\tan(\theta)\sqrt{\tan(\theta)^2 + 1}} d\theta \Bigg|_{x=1}^{x=\infty} \\ &= -\frac{\sinh^{-1}(x)}{x} + \int \frac{\sec(\theta)^2}{\tan(\theta)\sec(\theta)} d\theta \Bigg|_{x=1}^{x=\infty} \\ &= -\frac{\sinh^{-1}(x)}{x} + \int \csc(\theta) d\theta \Bigg|_{x=1}^{x=\infty} \\ &= -\frac{\sinh^{-1}(x)}{x} - \ln |\csc(\theta) + \cot(\theta)| \Bigg|_{x=1}^{x=\infty}. \end{aligned}$$

Now, we substitute back $\cot(\theta) = 1/x$ and $\csc(\theta) = \sqrt{x^2 + 1}/x$:

$$-\frac{\sinh^{-1}(x)}{x} - \ln \left(\frac{\sqrt{x^2 + 1} + 1}{x} \right) \Bigg|_{x=1}^{x=\infty}.$$

Now, we use the logarithmic definition for $\sinh^{-1}(x)$:

$$\begin{aligned} & -\frac{\ln(x + \sqrt{x^2 + 1})}{x} - \ln \left(\frac{\sqrt{x^2 + 1} + 1}{x} \right) \Bigg|_{x=1}^{x=\infty} \\ &= 0 - 0 + \ln(1 + \sqrt{2}) + \ln(1 + \sqrt{2}) \\ &= 2 \ln(1 + \sqrt{2}). \end{aligned}$$